# Medical Question Answering on PubMedQA: Analysing Text Representations and Classifier Design

**Group 16**
EMATM0067 – Introduction to AI and Text Analytics
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## Abstract

Biomedical question answering requires classifying whether published evidence supports a *yes*, *no*, or *maybe* answer to a clinical research question. We investigate this task on the PubMedQA expert-annotated subset, following the Jin et al. (2019) 500/500 evaluation protocol, with 10-fold stratified cross-validation for classical models and a validation-fold strategy for transformer tuning, along two design axes: (1) text representation (TF-IDF n-grams, BioBERT embeddings, varying input compositions) and (2) classifier design (traditional classifiers, zero-shot BART-MNLI, fine-tuned transformers with class-weighted, focal, ordinal and hierarchical objectives, and parameter-efficient LoRA). Our best configuration—PubMedBERT fine-tuned with focal loss and a validation-tuned *maybe* threshold—achieves test macro-F1 of 0.579 (73.2% accuracy), a statistically significant improvement over the baseline TF-IDF+LR pipeline (McNemar $p<0.001$). Per-class analysis confirms that *maybe* remains the central bottleneck: the best-overall model scores $F1_{\text{maybe}}=0.16$, while the best Q+Ctx+Ans BioBERT model reaches $F1_{\text{maybe}}=0.27$ at the cost of lower overall macro-F1, revealing a genuine trade-off between overall balance and minority-class recovery.

## 1 Introduction

Biomedical question answering (QA) aims to determine whether published scientific evidence supports a given clinical research question.[1] This capability matters for evidence-based medicine: the volume of biomedical literature grows faster than clinicians can manually review it (1), and automated QA can accelerate clinical decision support (2).

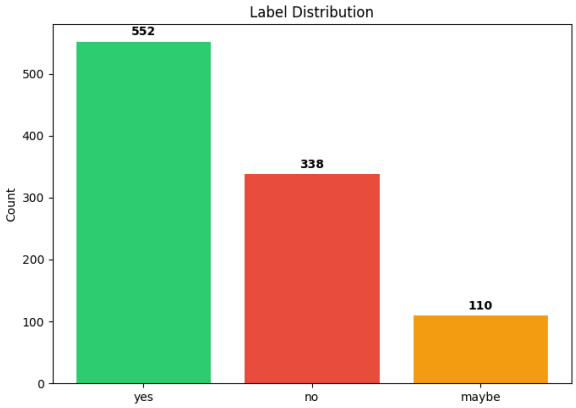


Figure 1: PubMedQA label distribution. The minority *maybe* class accounts for 11.0% of instances, creating a severe class imbalance that affects every downstream classifier.

The PubMedQA dataset (4) frames this problem as three-class classification. Given a research question and context paragraphs from a PubMed abstract, a system predicts whether the answer is *yes*, *no*, or *maybe*. The 1,000 expert-annotated instances are distributed as *yes* 552 (55.2%), *no* 338 (33.8%), and *maybe* 110 (11.0%), as shown in Figure 1. With only 11% of instances, *maybe* is severely under-represented.

The task is hard for three reasons. First, the skewed distribution causes classifiers to over-predict the majority class. Second, *maybe* captures epistemic uncertainty—evidence that is insufficient or inconclusive—rather than a midpoint between *yes* and *no*. Detecting it requires reasoning about evidence sufficiency, which is harder than recognising affirmative or negative lexical pat-

[1] Code repository: https://github.com/tanishkshindepatil01/ai-text-analytics-coursework-team16-task1. Project board: https://abhigyanxkashyap.atlassian.net/jira/software/projects/AT/boards/2.

terns. Third, the asymmetry between short questions (12.9 words on average) and long contexts (200.2 words on average of dense biomedical jargon) dilutes discriminative cues in bag-of-words models (5).

Exploratory analysis shows that surface features are weak: question lengths and context lengths overlap substantially across classes, so a classifier relying only on such features struggles to separate *yes*, *no*, and *maybe*. This motivates our later quality-prediction analysis, where handcrafted features provide only modest signal above chance.

**Central research question.** We design experiments to identify where the bottleneck in PubMedQA performance lies—input representation, classifier choice, or the difficulty of identifying uncertainty in the *maybe* class—and organise the investigation along two axes.

**Axis 1: Text Representation.** How do input composition (question only, question+context, question+context+answer) and encoding method (TF-IDF vs. BioBERT embeddings) affect classifier performance?

**Axis 2: Classifier Design.** How does classifier choice, from traditional models to fine-tuned transformers with specialised training objectives, affect handling of class imbalance and the *maybe*-class bottleneck?

**Contributions.**

1. A two-dimensional comparison of 20 model configurations spanning both axes, each motivated by a testable hypothesis and evaluated under the Jin et al. (2019) 500/500 protocol, using 10-fold stratified CV for classical models and a held-out validation fold for transformer tuning.

2. A decomposed analysis showing that input composition has little effect under TF-IDF ($\approx$ 0.35 macro-F1 across all variants), but representation and fine-tuning together lift BioBERT to 0.555 and PubMedBERT with focal loss to 0.576; a post-hoc *maybe* threshold adds a further 0.003 for a final 0.579.

3. A systematic *maybe*-class intervention comparison (threshold tuning, focal loss, hierarchical, LoRA, ordinal loss) that exposes a clear trade-off: *Q+Ctx+Ans* BioBERT achieves the highest *maybe* F1 (0.27) at lower overall macro-F1, while PubMedBERT focal+threshold wins overall macro-F1 (0.58) at lower *maybe* F1 (0.16).

4. Detailed error analysis with McNemar's test, pairwise Jaccard error overlap, K-Means cluster analysis, and representative misclassified examples, suggesting that failure on *maybe* stems from difficulty judging evidence sufficiency. A complementary quality predictor using handcrafted surface features alone performs near chance, consistent with this view.

### 1.1 Related Work

Biomedical QA has progressed from rule-based extraction (3) to transformer-based methods. BioBERT (2), pre-trained on PubMed, sets strong biomedical benchmarks; PubMedBERT (6) shows that pre-training from scratch on biomedical text can outperform continued pre-training from a general-domain checkpoint, while ClinicalBERT (5) reports similar gains from clinical notes. We use both BioBERT and PubMedBERT. Jin et al. (4) show that PubMedQA is especially challenging in its three-class form, with the *maybe* class being the hardest; our error analysis confirms this pattern. Related work includes transfer learning (8), retrieval augmentation (9), text-to-text generation (10), and sequence tagging (11). In contrast, we focus on a systematic comparison of text representation and classifier design, coupled with targeted *maybe*-class interventions.

## 2 Methods

Figure 2 overviews the experimental pipeline. The two design axes are text representation and classifier design, while preprocessing and evaluation are fixed.

### 2.1 Dataset and Preprocessing

We use the expert-annotated subset of PubMedQA (4) (1,000 instances). Following Jin et al.'s published evaluation protocol, we split the data into 500 cross-validation and 500 test samples using stratified sampling with seed 0, reconstructed programmatically. The 500 CV samples are further divided into 10 stratified folds for internal tuning, with fold 0 reserved as a development set for transformer hyperparameter and threshold selection. Table 1 summarises the splits. Each instance contains a research question, context paragraphs from a PubMed abstract, a long-form answer, and a label. We apply lowercasing, whitespace normalisation, and punctuation cleaning uniformly to all text fields.

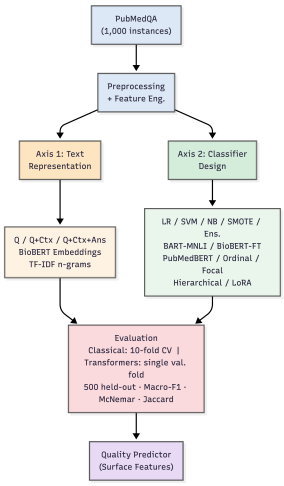


Figure 2: Experimental pipeline. Both axes feed a shared evaluation stage. Classical models use 10-fold CV; transformers use a single validation fold due to compute cost.

Table 1: PubMedQA splits (Jin et al. 2019 protocol). Stratified sampling preserves label proportions across both halves.

| Split | Total | Yes | No | Maybe |
|---|---|---|---|---|
| CV (50%) | 500 | 276 | 169 | 55 |
| Test (50%) | 500 | 276 | 169 | 55 |
| Full dataset: | 1,000 | 552 | 338 | 110 |

Labels are encoded as integers: *no*=0, *maybe*=1, *yes*=2.

### 2.2 Baseline System

The baseline uses TF-IDF (unigrams and bigrams, max 30,000 features, English stop-words) with a class-weighted Logistic Regression classifier. The model receives `question [SEP] context` as input. We additionally engineer 10 interpretable handcrafted features from question and context only—text lengths, number of context sentences, and binary indicators for negation, causation, comparison, association, risk, evidence strength (e.g. *significant, demonstrated*), and hedging (e.g. *may, suggest, inconclusive*)—reserved for the quality-predictor analysis. Long-answer-derived features are excluded there to prevent leakage.

### 2.3 Axis 1: Text Representation

We hypothesise that the information available to the classifier strongly affects performance. Five strategies test this:

**A1a (Question only).** Question text only. Expected to perform worst: answers require context.

**A1b (Q+Ctx).** The baseline composition. Tests whether evidence paragraphs aid classification.

**A1c (Q+Ctx+Ans, TF-IDF).** The long-answer field summarises the authors' interpretation; we test whether sparse TF-IDF can exploit it, given that signal may be diluted by the large biomedical vocabulary.

**A1d (BioBERT Embeddings + LR).** We extract 768-d BioBERT embeddings (2) from the Q+Ctx input using a biomedical sentence-transformer variant[2] and feed them to a class-weighted logistic regression. This tests whether dense semantic representations beat sparse vocabulary features on the same classifier.

**A1e (BioBERT fine-tuned on Q+Ctx+Ans).** Same classifier family as A2g below, but different input. We test whether transformers—unlike TF-IDF—can exploit the explicit author conclusions that the long answer contains.

**Caveat on Q+Ctx+Ans.** The long answer is a human-written summary that often contains explicit answer-indicative wording, so A1c and A1e should be interpreted as an enriched-input setting rather than pure question-over-abstract reasoning. We include it because it is a standard PubMedQA field and because it decomposes cleanly into an Axis-1 intervention.

### 2.4 Axis 2: Classifier Design

For the classical Axis-2 models we fix the base representation to combined word- and character-level TF-IDF on Q+Ctx ($\sim$70,000 features when used). For transformer-based Axis-2 models we use the enriched Q+Ctx+Ans input where stated. This separates classical classifier comparisons from transformer-specific and loss-function interventions.

**A2a–c (Tuned LR, calibrated SVM, Multinomial NB).** Inner 5-fold grids over $C \in \{0.05, \ldots, 5\}$ for LR/SVM and $\alpha \in \{0.05, \ldots, 1.0\}$ for NB. A2b additionally tunes a *maybe* decision threshold on the development fold after calibration.

**A2d (SMOTE+LR).** Applied *inside* each CV fold to avoid leakage. Tests whether synthetic minority oversampling recovers *maybe*.

[2] `pritamdeka/BioBERT-mnli-snli-scinli-scitail-mednli-stsb` on HuggingFace.

**A2e (Soft-voting ensemble).** LR + calibrated SVM + NB. Tests whether averaging weak TF-IDF classifiers helps.

**A2f (Zero-Shot BART-MNLI).** `facebook/bart-large-mnli` with hypothesis template *"The answer to this medical question is {label}"*. No training; probes off-the-shelf NLI transfer.

**A2g (Fine-tuned BioBERT, single config).** `dmis-lab/biobert-base-cased-v1.2` fine-tuned on Q+Ctx+Ans with class-weighted cross-entropy, lr $2\times10^{-5}$, max_len 384, 8 epochs, early-stopping patience 2:

$$\mathcal{L}_{\text{CE}} = -\sum_{i=1}^{N} w_{y_i} \log \hat{p}(y_i \,|\, x_i), \quad w_c = \frac{N}{K \cdot n_c} \tag{1}$$

with $N$ the total sample count, $K = 3$, and $n_c$ the count of class $c$.

**A2h (BioBERT HP-tuned).** Grid search over lr $\in \{1, 2, 3, 5\}\times10^{-5}$ and max_len $\in \{256, 384\}$, selected by development macro-F1.

**A2i (PubMedBERT fine-tuned).** `microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract`, pre-trained from scratch on PubMed abstracts. Tests whether stronger in-domain pre-training helps.

**A2i+Focal (PubMedBERT + Focal Loss).** We replace cross-entropy with class-weighted focal loss (7) ($\gamma = 2$), which down-weights easy examples so gradients focus on hard ones—expected to help with minority classes:

$$\mathcal{L}_{\text{focal}} = -\sum_{i} w_{y_i} (1 - p_{y_i})^{\gamma} \log p_{y_i}. \tag{2}$$

A post-hoc *maybe* threshold is then tuned on the development fold (A2i+Focal+Thr).

**A2j (BioBERT ordinal-aware loss).** The labels are ordered (*no* < *maybe* < *yes*) and the clinical cost of confusing *yes* with *no* is greater than confusing either with *maybe*. We add an MSE penalty on the expected ordinal position:

$$\mathcal{L}_{\text{ord}} = \mathcal{L}_{\text{CE}} + \alpha \cdot \mathbb{E}\big[(\hat{y}_{\text{ord}} - y_{\text{ord}})^2\big] \tag{3}$$

where $\hat{y}_{\text{ord}} = \sum_c c \cdot p(c \mid x)$ and $\alpha = 1.0$.

**A2j-Hier (Hierarchical).** Two-stage classifier: Stage 1 separates *maybe* from definitive (yes/no); Stage 2 classifies definitive samples as *yes* or *no*. Tests whether decomposing the problem helps.

Table 2: Experimental setup.

| Component | Detail |
|---|---|
| Primary metric | Macro-F1 |
| Secondary metrics | Accuracy, per-class F1 |
| CV strategy | 10-fold stratified on 500 CV set |
| Held-out test | 500 samples, evaluated once |
| Transformer dev fold | fold 0 ($\approx$ 50 samples) |
| Significance test | McNemar (paired) |
| Class weighting | Inverse-frequency (Eq. 1) |
| *Core software:* Scikit-learn, HuggingFace, imbalanced-learn, SentenceTransformers, PEFT | |

**A2k (LoRA PubMedBERT).** Parameter-efficient fine-tuning with LoRA (rank 8, $\alpha$=16, dropout 0.1, applied to query/value projections), trainable parameters reduced by $\sim$99.7%. Tests whether lightweight adaptation suffices.

### 2.5 Quality Predictor

We examine whether baseline correctness can be predicted from surface features alone. Using the 10 handcrafted features (Section 2.2), we train Random Forest, MLP, and Logistic Regression predictors on cross-validated correctness labels of the baseline and evaluate them on the held-out test set. This is deliberately an exploratory baseline analysis: if even weak classifiers produce errors predictable from shallow input features, that would motivate a simple input-based abstention policy; if not, confidence-based routing would be more appropriate.

## 3 Experimental Evaluation

### 3.1 Metrics and Setup

We use macro-F1 as the primary metric because the *maybe* class is severely under-represented and accuracy is misleading (a trivial all-*yes* classifier scores $\approx$ 55%). We additionally report per-class F1, accuracy, McNemar's test for paired significance, Jaccard error overlap between models, and K-Means cluster analysis. Table 2 summarises the setup. All scores in this section refer to the 500-sample held-out test set; CV scores, where reported, are the mean across the 10 stratified CV folds.

### 3.2 Baseline Results

The TF-IDF + LR baseline achieves 10-fold CV macro-F1 of 0.355 $\pm$ 0.057 and test macro-F1 of 0.356 (accuracy 51.2%)—below the 55% majority-class floor on accuracy. Figure 3 shows the characteristic failure mode: the classifier over-predicts *yes*

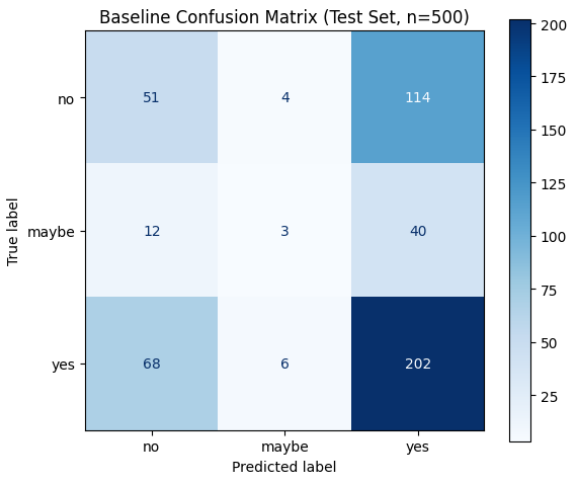


Figure 3: Baseline confusion matrix (TF-IDF + LR, test set, n=500). Only 3/55 *maybe* samples are correctly classified, illustrating the majority-class bias and the minority-class bottleneck that every later model must address.

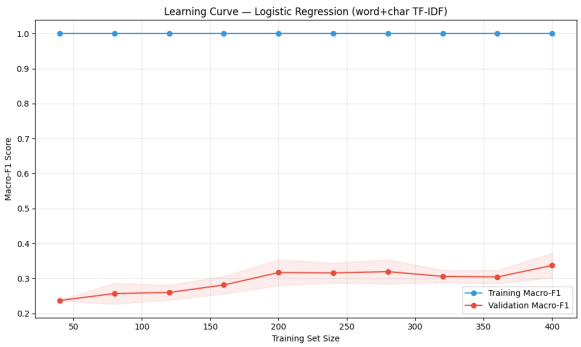


Figure 4: Learning curve for tuned Logistic Regression on word+char TF-IDF features. Training macro-F1 is $\approx$ 1.00 from the earliest fold size, while validation macro-F1 plateaus near 0.33. The persistent gap indicates that sparse features over-fit training n-grams rather than learning generalisable evidence-strength cues.

(356 predictions) and *no* (131), almost never predicts *maybe* (13), and only 3 of the 55 true *maybe* samples are recovered.

Figure 4 plots the baseline's learning curve: training macro-F1 saturates at 1.00 from fold size 40 upwards while validation macro-F1 drifts between 0.24 and 0.34, indicating severe over-fitting to training n-grams that do not generalise. This gap persists regardless of training-set size, consistent with a representation bottleneck rather than a data bottleneck.

### 3.3 Axis 1 Results: Text Representation

Under TF-IDF, input composition barely moves macro-F1: Q only 0.364, Q+Ctx 0.356, Q+Ctx+Ans 0.347. Adding context *or* the long answer does not help a sparse classifier—discriminative cues dilute against the large biomedical vocabulary. BioBERT embeddings + LR (A1d) on the same Q+Ctx input lift macro-F1 to 0.397 and, crucially, provide the first non-trivial *maybe* F1 (0.183), confirming that representation, not input composition, is the Axis-1 lever.

Table 3: Decomposing the gain from baseline to the best observed model.

| Configuration | Test F1 | $\Delta$ |
|---|---|---|
| TF-IDF + LR, Q+Ctx (baseline) | 0.356 | — |
| + Q+Ctx+Ans (still TF-IDF) | 0.347 | $-0.009$ |
| + BioBERT embeddings (still LR) | 0.397 | $+0.041$ |
| + BioBERT fine-tuned, Q+Ctx+Ans (A1e) | 0.555 | $+0.199$ |
| + PubMedBERT backbone + focal loss (A2i+Focal) | 0.576 | $+0.021$ |
| + Validation-tuned *maybe* threshold | **0.579** | $+0.003$ |

The decomposition in Table 3 traces the path from baseline to the best-observed model: the largest step comes from introducing a fine-tuned transformer on Q+Ctx+Ans (+0.198), followed by switching the backbone to PubMedBERT and adding focal loss (+0.022), followed by a final post-hoc *maybe* threshold (+0.003). Hyperparameter tuning of BioBERT reaches 0.502, $+0.146$ above baseline, showing that transformer performance is sensitive to optimisation choices; however, the strongest gains still come from combining enriched input, domain-adapted pre-training, and focal/threshold adjustment.

### 3.4 Axis 2 Results: Classifier Design

Three TF-IDF variants—tuned LR (A2a), calibrated SVM (A2b), Multinomial NB (A2c)—and SMOTE+LR (A2d) all cluster at 0.24–0.36 macro-F1, confirming that classifier choice on top of sparse features does not break the plateau. The calibrated SVM collapses to predicting almost only *yes* and scores the lowest *no* F1 (0.012), illustrating that calibration on top of a biased classifier does not fix the bias itself.

**SMOTE (A2d).** Applied inside each fold, SMOTE+LR scores CV 0.353 and test 0.357—statistically indistinguishable from baseline. Oversampling synthetic minority points in high-dimensional sparse space does not create linguistic variation that a bag-of-words classifier can exploit.

**Zero-shot BART-MNLI (A2f).** 0.070 macro-F1 and 11.2% accuracy, with hedged biomedical lan-

guage systematically mapped to NLI label *neutral*. Off-the-shelf NLI transfer is not viable for this task without fine-tuning.

**Fine-tuned transformers.** BioBERT with the default configuration (A2g) scores only 0.493—barely above the best TF-IDF classifier—showing that fine-tuning is sensitive to hyperparameters on a 450-example training split. HP-tuning lifts it to 0.502 (A2h). PubMedBERT (A2i) reaches 0.570 at 75.6% accuracy (the highest accuracy of any model), but its *maybe* F1 is only 0.108—a 19-prediction *maybe* bias. Adding focal loss (A2i+Focal) lifts macro-F1 to 0.576 and *maybe* F1 to 0.157, and the validation-tuned threshold (A2i+Focal+Thr) yields the final 0.579/0.160.

**Ordinal loss (A2j).** In our run the ordinal penalty collapses *maybe* F1 to 0.000, actually *increasing* *yes*↔*no* confusion from 82 (standard BioBERT FT) to 108, and final macro-F1 drops to 0.464. The MSE-on-expected-class term pushes probability mass toward the adjacent classes of whichever extreme the model prefers, but at $\alpha$=1 it penalises correct *maybe* predictions that retain any residual mass on *yes* or *no*, squeezing the middle class out rather than regularising it.

**Hierarchical (A2j-Hier).** Stage 1 learned essentially no uncertainty detector on our training folds—it predicted zero *maybe* on the test set—so all final predictions defaulted to Stage 2. Final macro-F1 = 0.473 and *maybe* F1 = 0.000. Decomposing the problem does not help when Stage 1 does not have enough *maybe* examples to learn the binary boundary.

**LoRA PubMedBERT (A2k).** 0.327 macro-F1 with only 0.27% trainable parameters ($\approx$ 297K of 109M). Far below full fine-tuning, but well above the zero-shot floor, showing the rank-8 adapters do learn partial task structure—just not enough for the small training set.

**BioBERT Q+Ctx+Ans (A1e).** Same classifier family as A2g but with the long answer added. Macro-F1 rises from 0.493 to 0.555 and *maybe* F1 from 0.025 to 0.273—the highest *maybe* F1 of any model we tested. This is the single biggest intervention for the minority class.

### 3.5 Overall Model Ranking

Table 4 shows the main ranking with per-class F1; the remaining weaker traditional variants are discussed in the text. Four tiers emerge: (1) transformers on Q+Ctx+Ans (0.55–0.58); (2) HP-tuned / ordinal / hierarchical transformer variants (0.46–0.50); (3) BioBERT-embedding or SMOTE baselines (0.36–0.40); (4) sparse classical, LoRA, and zero-shot ($\leq$0.34).

Figure 5: Per-class F1 for the main reported models in Table 4. The *maybe* column is near-zero for most sparse baselines, reaches 0.27 with BioBERT fine-tuning on Q+Ctx+Ans, and stays at 0.16 even for the overall-best PubMedBERT Focal+Threshold model.

### 3.6 Per-Class F1 Analysis

Figure 5 shows per-class F1 for the main reported models in Table 4, while the remaining weaker traditional variants are discussed in the text. The *maybe* column is near-zero for almost every TF-IDF model, confirming that it is the dominant performance bottleneck. Only five models reach *maybe* F1 $\geq$0.15, and only BioBERT Q+Ctx+Ans reaches $\geq$0.20. Importantly, the top-ranked model on macro-F1 (PubMedBERT Focal+Thr) does *not* have the best *maybe* F1; that belongs to BioBERT Q+Ctx+Ans. This is the central trade-off of the study.

### 3.7 Error Analysis

#### 3.7.1 Class Confusion Patterns

Figure 6 shows per-class error rates for the best-overall model (A2i+Focal+Thr): *yes* 17.0%, *no* 23.7%, *maybe* 85.5%. Figure 7 decomposes the errors: of the 55 true *maybe* samples, 8 are recovered, 33 are misclassified as *yes*, and 14 as *no*. This is a 4$\times$ improvement over the baseline's 3/55, but *maybe* recall is still the dominant residual error mode. Yes↔no reversals total 50 cases (26 *yes*→*no*, 24 *no*→*yes*)—the clinically most serious

Table 4: Main results on the 500-sample test set, sorted by macro-F1. Weaker traditional variants (A2a, A2b, A2d, A2e) are discussed in the text but omitted here for brevity. Bold indicates the best value per column.

| Label | Model | Acc | F1 | $F1_{no}$ | $F1_{maybe}$ | $F1_{yes}$ |
|---|---|---|---|---|---|---|
| A2i+Focal+Thr | PubMedBERT Focal+Threshold | 0.732 | **0.579** | 0.763 | 0.160 | 0.815 |
| A2i+Focal | PubMedBERT Focal | 0.728 | 0.576 | 0.756 | 0.157 | 0.815 |
| A2i | PubMedBERT FT | **0.756** | 0.570 | **0.770** | 0.108 | **0.832** |
| A1e | BioBERT Q+Ctx+Ans | 0.666 | 0.555 | 0.633 | **0.273** | 0.759 |
| A2h | BioBERT HP-Tuned | 0.594 | 0.502 | 0.653 | 0.190 | 0.662 |
| A2g | BioBERT FT (default config) | 0.682 | 0.493 | 0.683 | 0.025 | 0.768 |
| A1e-Thr | BioBERT Q+Ctx+Ans + Threshold | 0.680 | 0.490 | 0.642 | 0.060 | 0.768 |
| A2j-Hier | Hierarchical | 0.690 | 0.473 | 0.640 | 0.000 | 0.777 |
| A2j | BioBERT Ordinal Loss | 0.672 | 0.464 | 0.636 | 0.000 | 0.757 |
| A1d | BioBERT Embeddings + LR | 0.490 | 0.397 | 0.400 | 0.183 | 0.607 |
| A1a | Q-only (TF-IDF + LR) | 0.512 | 0.364 | 0.408 | 0.061 | 0.623 |
| A1b | Baseline (Q+Ctx, TF-IDF+LR) | 0.512 | 0.356 | 0.340 | 0.088 | 0.639 |
| A1c | Q+Ctx+Ans (TF-IDF + LR) | 0.510 | 0.347 | 0.347 | 0.058 | 0.637 |
| A2c | Multinomial NB (word+char) | 0.548 | 0.337 | 0.335 | 0.000 | 0.676 |
| A2k | LoRA PubMedBERT | 0.406 | 0.327 | 0.428 | 0.099 | 0.453 |
| A2f | Zero-Shot BART-MNLI | 0.112 | 0.070 | 0.012 | 0.199 | 0.000 |

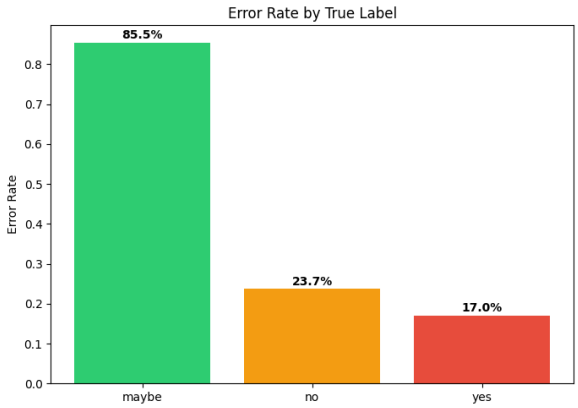


Figure 6: Per-class error rates for the best model (A2i+Focal+Thr): 85.5% on *maybe* versus 17.0%/23.7% on the majority classes. Uncertainty remains the central bottleneck.

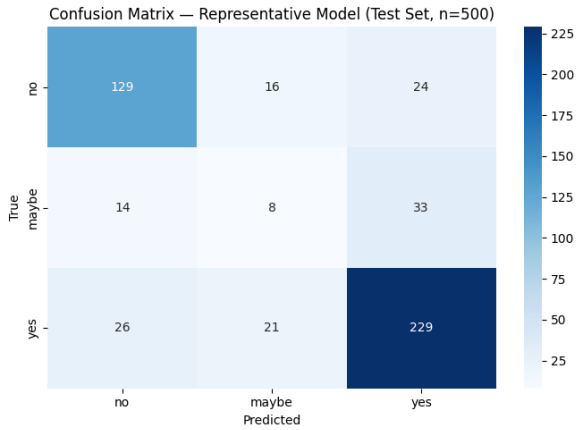


Figure 7: Confusion matrix of the best model (A2i+Focal+Thr, test set, n=500). Of 55 true *maybe* samples, 8 are recovered; 33 are pushed to *yes* and 14 to *no*. Yes↔no reversals total 50 cases.

error class because it inverts the recommendation rather than merely adding caution.

### 3.7.2 Statistical Significance

McNemar's paired test on the 500-sample test set yields 154 cases in which the focal+threshold model is correct and the baseline is wrong, against 44 cases in the opposite direction ($b$=154, $c$=44, $p$<0.001). The improvement over the baseline is statistically significant.

### 3.7.3 Representative Misclassified Examples

Table 5 illustrates the four dominant confusion patterns. The *yes→no* and *no→yes* cases are direction-of-evidence mistakes where the wording contains cues that point the opposite way. The *maybe→yes/no* cases show the model forcing ambiguous evidence to a conclusive label; this happens despite the focal loss, which suggests the issue is not gradient weighting alone but genuinely poor

Table 5: Representative misclassified examples from A2i+Focal+Thr (shortened).

| Pattern | Question (shortened) | True | Pred |
|---|---|---|---|
| yes→no | Discordance in RBC morphology reporting? | yes | no |
| no→yes | Covering resection margin after distal pancreatectomy? | no | yes |
| maybe→yes | Pedestrian street-crossing accuracy by vision status? | maybe | yes |
| maybe→no | Web vs clinic HIV+ symptom differences? | maybe | no |

evidence-sufficiency signals.

These examples show that the main failures are not simple lexical mistakes: *yes*/*no* errors reverse the direction of evidence, while *maybe* errors occur when the model forces inconclusive evidence into a definitive answer.

#### 3.7.4 Cross-Model Error Overlap

Figure 8 shows pairwise Jaccard similarity between error sets of the top-4 models. The two PubMedBERT-backbone variants (A2i+Focal, A2i+Focal+Thr) share 97% of their errors—the threshold is a shallow post-hoc adjustment. A2i PubMedBERT-FT shares 68–70% with them. In contrast, A1e BioBERT Q+Ctx+Ans shares only 40–44% with the PubMedBERT cluster, indicating genuinely different failure modes. 74 test examples (14.8%) are misclassified by all four top models—a plausible floor on irreducible difficulty for current approaches on this split. The low BioBERT/PubMedBERT overlap suggests an ensemble of A1e and A2i+Focal would combine complementary strengths (maybe-F1 from A1e, macro-F1 from A2i+Focal) and is a natural direction for future work.

#### 3.7.5 Cluster-Based Error Analysis

K-Means on the BioBERT sentence embeddings of the test set yields 4 semantic clusters whose macro-F1 for the best model ranges from 0.539 (cluster 2, 150 samples, 14 *maybe*) to 0.609 (cluster 0, 58 samples, 6 *maybe*, 0.00 *maybe* F1). Cluster 1 (125 samples, 17 *maybe*) achieves the strongest *maybe* F1 (0.267). The performance variation across clusters indicates that some semantic regions of the embedding space are much more uncertainty-heavy than others, and that per-cluster decision rules could, in principle, lift minority-class recovery further.

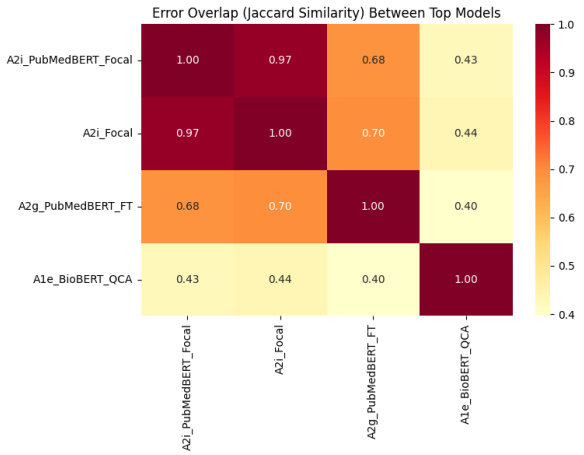


Figure 8: Jaccard error overlap between the top-4 models. The two PubMedBERT-focal variants share 97% of errors; A1e BioBERT Q+Ctx+Ans shares only 40–44% with the PubMedBERT cluster, indicating genuinely complementary failure modes.

### 3.8 Quality Predictor Results

Three surface-feature predictors over the 10 handcrafted features perform weakly: RF reaches 0.514 macro-F1, LR reaches 0.498, and MLP reaches 0.403 for predicting baseline correctness. Thus, only the RF model is marginally above the $\approx 0.50$ majority-correct chance floor, while the neural baseline underperforms. Feature importance (RF) ranks context length and number of context sentences at the top, with hedging and negation contributing little. This means baseline errors are *not* well predicted by shallow input characteristics: longer or denser inputs correlate weakly with difficulty, but that is a blunt instrument. In line with our Axis-1 finding, meaningful error prediction likely requires model-internal uncertainty signals (softmax entropy, margin) rather than purely input-based features. For deployment, this argues for routing low-confidence model outputs to human review rather than abstaining on input-length heuristics.

## 4 Discussion

**Why Q+Ctx+Ans helps transformers but not TF-IDF.** The long answer often condenses the authors' interpretation—markers like *"our findings suggest"*, *"associated with"*, *"may be safe"*, or *"further investigation is warranted"* are discourse cues about evidence sufficiency. TF-IDF treats these as lexical units alongside domain vocabulary and they are diluted; an attention-based encoder can condition them on context. This single in-

put change lifts BioBERT macro-F1 from 0.493 (A2g) to 0.555 (A1e) and *maybe* F1 from 0.025 to 0.273. Biomedical NLP systems that assume abstracts carry all practical signal likely leave this lift on the table.

**Backbone choice matters more than loss function.** Under the transformer setting, switching the backbone from BioBERT to PubMedBERT lifts macro-F1 by approximately 0.07. Adding focal loss on top of PubMedBERT gives a smaller lift from 0.570 to 0.576, while the post-hoc threshold adds a further $\approx 0.003$. In contrast, loss-only interventions on BioBERT (ordinal, hierarchical) produced worse, not better, minority-class recovery on this split. Loss engineering is not a substitute for a better-adapted pre-training distribution.

**The overall vs. minority trade-off is genuine.** No single model dominates all metrics. PubMedBERT Focal+Thr wins overall macro-F1; PubMedBERT FT wins accuracy alone (75.6%); BioBERT Q+Ctx+Ans wins *maybe* F1. For a deployment that prioritises not missing uncertain evidence (e.g. flagging cases for clinician review), A1e is preferable even though its macro-F1 is 0.024 lower. The 40–44% Jaccard overlap between A1e and the PubMedBERT cluster suggests an ensemble may capture both strengths.

**Why our ordinal loss did not help this run.** Prior work reports ordinal penalties reducing severe-class confusion, but at $\alpha$=1.0 and with only around 50 *maybe* samples in the transformer training folds our model learned to collapse *maybe* to zero predictions. The MSE term penalises any residual probability mass on the extreme classes, which under inverse-frequency weighting and a sparse minority effectively forbids the model from ever preferring *maybe* over the majority classes. A smaller $\alpha$, a clipped distance penalty, threshold tuning, or combining ordinal loss with focal-style reweighting may recover the intended behaviour.

**Practical trade-offs.** Sparse TF-IDF baselines train in seconds on CPU; full transformer fine-tuning requires a GPU and several minutes per epoch per configuration. For exploratory research, A2i+Focal+Thr's $+0.22$ macro-F1 over baseline clearly justifies the cost. In deployment, resource constraints mean a system may need to compromise between predictive performance, minority-class recovery, and compute budget, with LoRA as a middle point.

### 4.1 Limitations

*Dataset constraints.* Our 500/500 split contains only 55 *maybe* samples in each half, which limits both training signal and the precision of per-class evaluation. Protocol-consistent programmatic reconstruction rather than loading the exact published PMID list means our split is not bit-identical to Jin et al.'s.

*Evaluation constraints.* Classical models use full 10-fold CV; transformers use a single fold-0 validation split due to GPU cost, so their tuning stability is less well-estimated. Because many variants are compared on the same held-out test set, the ranking should be read descriptively rather than as a strict deployment-style selection.

*Methodological constraints.* The hyperparameter search covered 8 configurations across 2 axes; larger budgets may yield further gains. The ordinal and hierarchical variants were only tested with one Q+Ctx+Ans setting; we did not sweep $\alpha$ values, Stage-1 thresholds, or alternative hierarchical decision rules. Calibration analysis (ECE, reliability diagrams, temperature scaling) would also be a useful complement but is left for future work, since a rigorous comparison across multiple models was outside our compute budget.

*Input-composition caveat.* Because Q+Ctx+Ans includes the human-written long answer—a summary that often contains explicit answer-indicative language—the enriched-input transformer results (A1c, A1e, A2i and variants) should not be interpreted as pure abstract-only biomedical question answering. They represent an upper bound on what a system can achieve when author conclusions are available at inference time.

### 4.2 Future Work

Natural next steps: (i) an ensemble of A1e and A2i+Focal, motivated by their low 40–44% error overlap; (ii) semi-supervised learning on the 211k unlabelled PubMedQA examples; (iii) temperature scaling / MC-dropout calibration for clinical deployment; (iv) testing focal loss specifically on the BioBERT Q+Ctx+Ans setup, since focal loss was reported only for the PubMedBERT variant; (v) abstention based on softmax margin, routing low-confidence cases to human review.

## 5 Conclusions

On PubMedQA, we evaluated 20 model configurations across two design axes—text representation

and classifier design—under the Jin et al. (2019) 500/500 protocol, using 10-fold stratified CV for classical models and a held-out validation fold for transformer tuning. Three findings emerge. **First,** input composition interacts with model capacity: adding the long answer does nothing under TF-IDF ($\approx 0.35$) but takes fine-tuned BioBERT from 0.493 to 0.555 and *maybe* F1 from 0.025 to 0.273. The biggest single lever for the minority class is input, not loss. **Second,** the overall-best model—PubMedBERT + Focal Loss + validation-tuned *maybe* threshold—reaches 0.579 macro-F1 (73.2% accuracy) on the 500-sample test set, a statistically significant improvement over baseline (McNemar $p<0.001$), but its *maybe* F1 is 0.160, below A1e's 0.273. There is no free lunch: overall balance and minority recovery trade off. **Third,** *maybe* is the residual bottleneck with an 85.5% error rate even for the best model. This suggests the task needs evidence-sufficiency reasoning that neither bag-of-words features nor current transformer losses fully capture; the quality-predictor result (surface features barely above chance) supports this view. Taken together, our results argue that safe deployment of medical QA systems should look beyond headline macro-F1: error distribution (yes↔no reversals vs. uncertainty misses), minority-class recovery, and cross-model error overlap matter at least as much.

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